Blinded-V5: the LGI-based compute module

design rationale, construction, parameters, and measurements

local-mixing project

2026-09-04

Where it fits in the pipeline

A secret circuit C on n wires is protected in two steps: it is first **gadgetized** into an equivalent but structurally-scrambled circuit on $2n$ wires, which is then **mixed** (fmix). Blinded-V5 lives inside the gadgetizer, and this document assumes you know the earlier gadgetize designs but nothing about V5.

The **gadgetize module** is a **5-stage** pipeline, each stage on $2n$ wires (data $0..n$, band/junk $n..2n$):

1. **slice** — a junk-guard zero-slice keyed on the band (dead at the input port, where the band is 0);
2. **seed the band wires** — each band wire set to $x_i \wedge \neg x_j$ from the honest inputs;
3. **compute** — realise the input circuit A on the $2n$ wires so the data half carries A's output and the band is only read;
4. **re-seed / re-randomise the band wires** — band turnover; a *separate* module emitted after the compute (the compute's own rerand bursts are masking hygiene and do not discharge it), so the band is junk at both ports;
5. **re-slice** — the closing junk-guard slice (fires at the output port).

One term to keep straight: the circuit A that the gadgetizer computes is **not C itself**. It is the **sliced sandwich of C** — the output of a *separate* **sandwich module** that wraps C (with a random D, interleaved slice blocks, a floated column) into a $2n$-wire circuit with $A(x, 0) = (junk, C(x))$. The sandwich *generates* the circuit that the gadgetize module then computes and mixes; it is a different module and is not one of the five stages above.

Blinded-V5 is a drop-in replacement for stage 3 — the compute. Where the earlier compute stages (the drip `route_fire`, the product-share carriers) realise A by *routing* each operand into position and firing a gadget there, Blinded-V5 instead embeds A **inside one enormous masking identity**: the compute's body is a cloud of long masking identities on the band, and A's gates fire from *within* the mask by a momentary, reversible unmasking of their operands. It takes A and emits an equivalent A' on the $2n$ wires; the other four stages are unchanged. (V5 re-randomises the band *internally* as part of its own masking hygiene — §2 — which is distinct from the separate stage-4 re-seed.)

Source: `src/preprocessing/blinded_v5.rs`. Driver: `gen_sandwich_gadget ... blinded-v5`; pipeline: `gss_mix.sh --gadgetization-mode blinded-v5`. The Markdown twin of this document is `BLINDED_V5_LGI_DESIGN.md`.

1 Why this design — the rationale

The compute stage must move the intermediate state as far as possible from the plaintext computation A while keeping A' exactly equivalent to A on the data wires. The design is a chain of forced moves.

(a) Long identities are the only thing that moves the state. Across the HD and affine heatmaps, short local edits barely displace the state; the ridge of recoverable structure only recedes when the masking identity woven through a wire is *long* — comparable to the circuit itself.

(b) The only long identity that moves the state is the commutative structure of an identity with a single active wire. A run of gates all targeting one active wire w and reading only controls is an identity iff the increments XOR to zero, and *all its gates commute* — which lets it be arbitrarily long and freely interleaved.

(c) Entangle identities on different active wires by sharing the control wires — the band. Every identity targets a data wire and reads only the band $(n..2n)$, so identities on different active wires also commute; a band wire participates in the masks of many data wires at once.

(d) Add random updates of the control wires from the active wires — in bursts. Band updates **re-randomise** the controls (fresh band entropy throughout the run, not just at the seed) and **break naive commutation-back** (once a band wire is updated mid-run, the identities reading it before and after no longer commute past it), without ruining correctness (each straddling identity is closed before the update or re-derived across it). The updates are emitted in **bursts**: one *slot* is $F \approx 8$K gates all targeting a single band wire b, each control drawn independently as a live-data or a band wire (so a burst spans $b \oplus\!= \mathit{data} \wedge \mathit{aux}$, $b \oplus\!= \mathit{aux} \wedge \mathit{aux}$, and $b \oplus\!= \mathit{data} \wedge \mathit{data}$). Bursting **concentrates** the band mixing into few slots (so fewer straddle-closures interfere with the LGIs) and gives each band wire a **data-wire-like burst-of-activity signature**.

(e) Embed A inside one circuit-spanning identity via hidden unmasking. Each gate fires from inside the mask: its operands are momentarily and reversibly unmasked into a linear combination of band wires (never bare), it fires as an expansion over those masked wires, then re-masks.

That is the whole design: a long, entangled, self-updating masking cloud with A firing from inside it. Everything in §2 is *how* to realise (a)–(e) as a single correct circuit.

2 The construction

A has n wires and m gates; A' has $2n$. Let u_w be the uses of wire w in A and w_w its writes.

Masking atom. $\mathsf{g57}(w, x, y) = w \oplus\!= 1 \oplus (\neg x \wedge y)$ (data target, band controls), i.e. the mask term $1 \oplus y \oplus xy$. A disjoint-pair LGI $\bigoplus_i \mathsf{g57}(w, cy_{2i}, cy_{2i+1})$ is a deg-2 mask; a $\mathsf{g57}$ and its reverse **linearise**, $\mathsf{g57}(w, r_1, r_2) \oplus \mathsf{g57}(w, r_2, r_1) = w \oplus r_1 \oplus r_2$ — the original read exploited this; the current read (**quad-fire**, step 2 below, default since 2026-09-06) deliberately does *not*, because a linearised operand is exactly affine in band wires for a window that the pipeline's reordering stages stretch into a leak (§5.0). Disjoint pairs, not a K-cycle (which would telescope to a bare operand); $K \geq 2$.

Why one pass — co-sampling. The masks, the band updates, and A's gates are **co-sampled**: produced together in a single forward pass, rather than laying a fixed masking scaffold first and threading A through it afterwards. The reason is hidden firing (§3): every A-gate must be straddled by a *real*, rerand-protected masking identity opened at the exact instant the gate fires. Only a co-sampled pass can open that identity on demand, so the firing-hiding is complete while reusing the wire's existing mask budget — at no size cost. (A fixed scaffold, filled in afterwards, starves at the tail and covers only 60–70% of gates; §3.)

The algorithm. A cheap setup, then a forward pass.

Setup.

1. **Mask budget.** Give each active wire w exactly $u_w + 1$ masking identities, split into w_w **straddle opens** (one per write to w, opened *on demand* at the gate that writes it — for hidden firing) and $\text{reads}_w + 1$ **filler opens** (opened between gates to keep w masked while it is read). At most `max_open` are open on a wire at once. This is the same budget a fixed scaffold would use, so the statistics are unchanged.
2. **Order.** Build the data-hazard DAG of A (`compute_deps`): a read is ordered after every earlier write of the wire, and a write after every earlier read — but same-target XOR writes commute, so there is *no* write-after-write edge. Any linear extension preserves A's function; a Kahn **ready queue** holds the gates whose predecessors are all placed.
3. **Rerand plan.** A shuffled list of `straddle_slots` + `repair_slots` band-update *slots* (§1d; auto `straddle_slots` $\approx m/(4K)$, `repair_slots` $= 0$).
4. **Rate.** Fire one rerand slot every *total_steps*/#*slots* primitive steps (a *step* = one A-gate placement or one filler open; *total_steps* $= m + \Sigma\,\text{filler}$), so the slots spread evenly and the last lands *during* the pass — no end-flush (a surplus of requested slots over *total_steps* warns rather than truncating silently).

Forward pass — repeat until all m gates are placed:

1. **Pop** a ready A-gate g (target wire c).
2. **Masked read (quad-fire).** The operand is only *read*, under whatever masks it currently carries: a wire under masks $m_i = 1 \oplus y_i \oplus x_i y_i \oplus z_i$ holds $w' = w \oplus \sum m_i$, i.e. $w = w' \oplus c_w \oplus s_w \oplus Q_w$ with the **constant** c_w (one per open `g57` pair, plus one for a negative literal), the **linear** band sum $s_w = \sum(y_i \oplus z_i)$ and the **quadratic** part $Q_w = \sum x_i y_i$. A thin operand is first brought up to `min_open` real masks (`ensure_min_open`, §5.6), so no wire is ever read bare or under a single mask; there are no read-time top-ups to undo any more. (Legacy linear read, `BV5_QUAD_FIRE=0`: complete each open `g57` with its reverse so the wire carries *operand* $\oplus\,\rho$, ρ a linear XOR of band wires, top up with pairs until $|\rho| \geq$ `min_mask`, undo the reverses after the fire — see §5.0 for why it was replaced.)
3. **Fire from inside the mask — two-control gates only, no clean ancillas (§5.8).** The fire is the product of the two operand polynomials. Expanding it into monomials would emit gates of degree up to 4, which the **frozen store cannot digest**: it is a ball of 2-control `g57` identities, so a wider gate is never spliced and would carry C's structure through the

whole mixing pipeline verbatim. The product is therefore realised with 2-control gates only, using **borrowed dirty wires** for the partial products — never clean ancillas, which would be 0 at every instant outside a fire, a function-level invariant that no rewriting can hide and whose non-zero stretches delimit the fire blocks. The identity is $t \mathrel{\hat{=}} h{\wedge}y$; $h \mathrel{\hat{=}} P{\wedge}x$; $t \mathrel{\hat{=}} h{\wedge}y$; $h \mathrel{\hat{=}} P{\wedge}x$, whose net effect is $t \mathrel{\hat{=}} P{\cdot}x{\cdot}y$ for *any* prior value of h ($h_0y \oplus (h_0{\oplus}Px)y = Pxy$), leaving h restored; the prior value also blinds the intermediate for free. A degree-4 term uses the same telescoping trick on two borrowed wires in 8 gates, and the s sums are expanded term by term so no mask aggregate is ever materialised. Four rules keep the borrowing safe, each found by measurement: draw the ancilla **per term**, excluding the target, the operands and the term's own wires; never a wire whose **own open masks contain the pair being XORed**; **cross-operand** groupings checked against the pairs open on any data wire, so no ancilla holds a mask's own quadratic term; and draw from the **band** (`BV5_ANC_POOL=all` also lends data wires, but a product XORed onto a masked data wire can partially cancel that wire's mask).

4. **Straddle (hidden firing, §3).** Split the monomial batch into two halves, shuffle each **independently** (they all target c and commute), emit the first half, **open a fresh lgi on c here** (one of its straddle opens, generated on demand), then emit the second half. The mid-fire open makes the module's net XOR on c equal $\Delta \oplus$ secret-mask, not the bare increment Δ. The whole block is **bracketed by a temporary cover mask on c** drawn away from every band wire of the operands' polynomials (opened before the first monomial, closed after the last; the straddle open is drawn away from those wires too): a band wire shared between one of c's masks and a fire monomial would cancel or fold that mask's uniform term and leave the mid-fire segments biased toward c_{new} (§5.7; $\sim$6 gates per fire, $\approx$3%).
5. **Undo** — emit the top-up gates again (each is an involution), restoring the operands' original mask set.
6. **Maybe rerand** — if the calibrated rate says so, emit one band-update slot (below).
7. **Release & fill** — decrement the in-degree of g's dependents, moving any that reach zero into the ready queue; then open a few **filler** LGIs on wires chosen by remaining filler budget.

Finish. Drain any remaining filler opens, then close every still-open LGI so the data wires end holding A's true output.

The rerand slot (step 6) is a §1d burst: one band wire b, $F \approx 8$K updates. A STRADDLE slot first **closes** the masks reading b (this is what thins masking past the ≈ 1024 knee — hence keeping `straddle_slots` at that budget); a REPAIR slot brackets the burst with each such mask's b-reading `g57` (old b cancels, new b re-adds, so the mask stays open — no thinning). Running in the same pass, the band updates protect the straddle opens automatically. **Coverage across the burst is a rule, not a statistic:** when the masks reading b are *all* of a wire's open masks, the slot first opens a *replacement* LGI on that wire (sampled away from b) and only then closes them — same-target XOR writes commute, so open-then-close never leaves an instant with no mask on the wire. Without it the wire sat bare, holding its plaintext value, until its next filler (§5.6: every interior bare interval of the earlier builds came from this — $\sim$450 per build, median $\sim 10^4$ gates). The same guard keeps every wire at `min_open` (2) masks, not just one: a burst opens as many replacements as needed before its closes, a filler or straddle open on a thin wire is followed by further opens, and a read on a thin operand keeps that many of its top-ups as real masks (§4). Every LGI sample (fillers, straddle opens, replacements, read top-ups) also draws its band wires

disjoint from every band wire already used by the wire's open masks. Any shared wire biases the mask sum: two identical pairs cancel ($1 \oplus y \oplus xy$ twice is 0 — the wire is functionally bare while the bookkeeping counts two masks), two identical balancing wires cancel ($z \oplus z = 0$, no uniform term left), and a balancing wire equal to another open pair's wire folds the linear and the quadratic term into an OR ($x \oplus \neg x \wedge y = x \vee y$, biased 3:1). Each of these showed up as a phi ≈0.25 segment population in the gauntlet at 32 band wires (§5.7); at 256 the last one still hits ∼10% of opens. The same discipline applies to the **burst gates**: a burst $b \mathrel{\hat{=}} lit(c_1) \wedge lit(c_2)$ with a data control reads that wire under its masks, so its other control is drawn away from that wire's mask wires (and two data controls may not share a mask wire) — otherwise the product strips a mask's uniform term and the burst's flip correlates with the plaintext (phi 0.14–0.26, seen at `max_open` 2, §5.7).

Correctness is verified exhaustively over all 2^n inputs × many band settings for $n \leq 6$, $K \in 2..5$, all `max_open` and both rerand kinds (`scratchpad/v6`: 891 gadgets, 0 mismatches, plus a 360-case reorder-preserves-A check; the linear read), by the unit test in `blinded_v5.rs` ($n = 6$, all 2^n inputs × 8 band states × 6 seeds, **both read modes**), and end-to-end in `gen_sandwich_gadget` (forward + reverse-honesty sample-verify PASS in both modes).

3 Hidden firing

An *atomic*, mask-restoring gate module would leak which gate fired: the active wire's before/after XOR across it equals the true increment $\Delta = comp \oplus \mathrm{lit}(a) \wedge \mathrm{lit}(b)$. That per-module increment is a robust invariant that survives the downstream mixing, so an adversary who segments the mixed circuit at module boundaries reads off A's gate list.

The fix uses the abundant masking gates on the *same* active wire: split the fire and emit one of c's LGI **opens between the halves**. It toggles c mid-fire and stays open past the module, so the net XOR on c is $\Delta \oplus$ (secret band-mask) — never the bare increment. Because everything is co-sampled, the straddling LGI is a real, rerand-protected scaffold LGI opened exactly when the gate is placed, so **every** gate is covered, reusing the wire's $u_w + 1$ budget at **no size cost** (measured: 7920/7920 gates on the n=128 sandwich). Earlier attempts — A-gate weaving (∼23%), an injected firing mask (100% but +43% size), straddling fixed-scaffold slots (60–70%, tail slot-starved) — are superseded by generating the slot on demand.

4 Parameters and why

knob	prod.	meaning	why this value
K	**2**	band wires per LGI = per-LGI **mask width** ($\lfloor K/2 \rfloor$ pairs; $\lvert\rho\rvert \approx$ `max_open` $\cdot K$) — not the temporal length	affine/deg-2-neutral (§5); smallest is best ∼linear, read cost qu $\lvert\rho\rvert$, so large K explo wastes a wire.
`max_open`	**3**	rolling cap on simultaneously-open LGIs per wire	wider ρ = more loca but read cost is quad 3 is the knee.
`quad_fire`	**on** (2026-09-06; two-control emission since 2026-09-07)	read operands from inside their quadratic masks, never linearise; emit the product through scratch wires as 2-control gates (§2 step 3, §5.8)	the linearised read le operand exactly affin wires for a window t reordering stage stre the C-vs-G ridge (§5 quad-fire has no win keeps the ridge at th through the whole p wide-monomial emis used was undigestibl store (37% of gates a controls); the scratch product is 2-control `BV5_QUAD_FIRE=0` =
`balanced`	**on** (2026-09-07)	every LGI (and every read top-up) adds one CNOT $w \mathrel{\hat{=}} z$ from a fresh band wire, so each mask term is $z \oplus 1 \oplus \neg x \wedge y$ — unbiased, still quadratic; band seed $x_i \oplus x_j$	a bare `g57` term is 1 in four, so a wire un open mask is *linearly* with its plaintext (p the C gate's firing p §5.6). Balanced mas channel to the null f (median phi 0.08) at gates (K=2, `max_ope` polynomial 8×8 → 1 monomials); it is wh the gauntlet's w1/w2 `BV5_BALANCED=0` = p `max_open` 2 balanced +10% variant, weake two-feature scans.
`min_open`	**2** (2026-09-07)	minimum open masks per data wire at *every* instant between its first and last mask; enforced wherever the count can drop or start low (burst replacements, filler/straddle opens, read top-ups kept)	one open mask is on term, which one visi monomial cancels (§ the rule 26.5% of cov wire-time (3% on the half; 2,398 interior o stretches, median 10 sat at one mask. At and no interior stret +0.03% gates (extra offset by reads that top-up). Must be <
`min_mask`	**auto** = `max_open` = 3	**hard floor** per operand read: quadratic mask terms	guarantees no operan read under fewer tha even in a rare draw

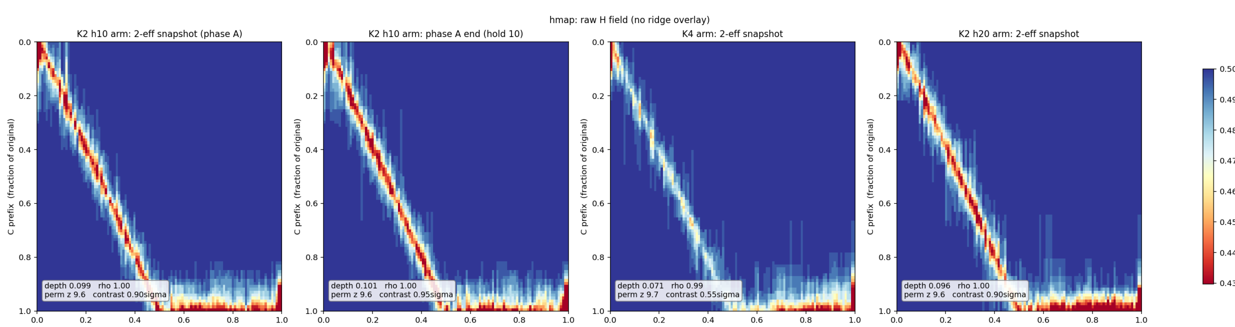
K sets each LGI's mask *width*, not its temporal length — the identity's circuit-spanning length is a property of the whole scaffold (§1e).

5 Measurements

All measurements are on the current co-sampled build with **burst rerand + the masking floor** (n=128, $|C| = |D| = 3000$, K=2). They match the pre-firing-fix baseline to within noise — neither the hidden-firing redesign nor the burst/floor refinements moved them (the burst change touches only the band wires, so it cannot move the data-wire ridge): degree-1 meanH = 0.4966, degree-2 meanH = 0.4967, exposed-$C < 0.35 \approx 5.6$–5.7%.

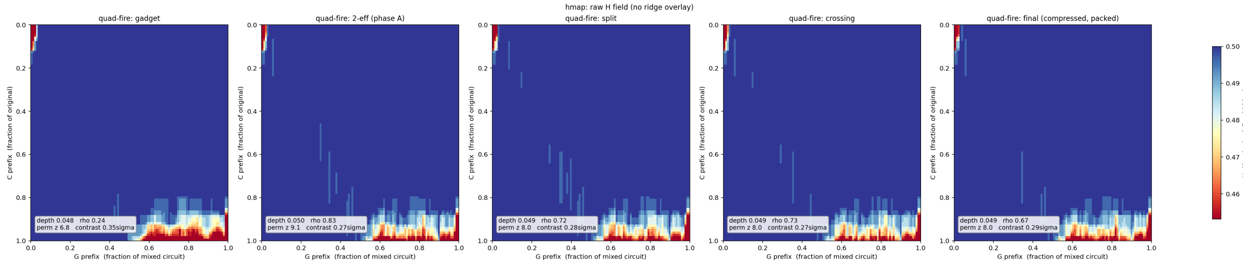
5.0 The affine ridge through the pipeline, and why the read mode changed (2026-09-06)

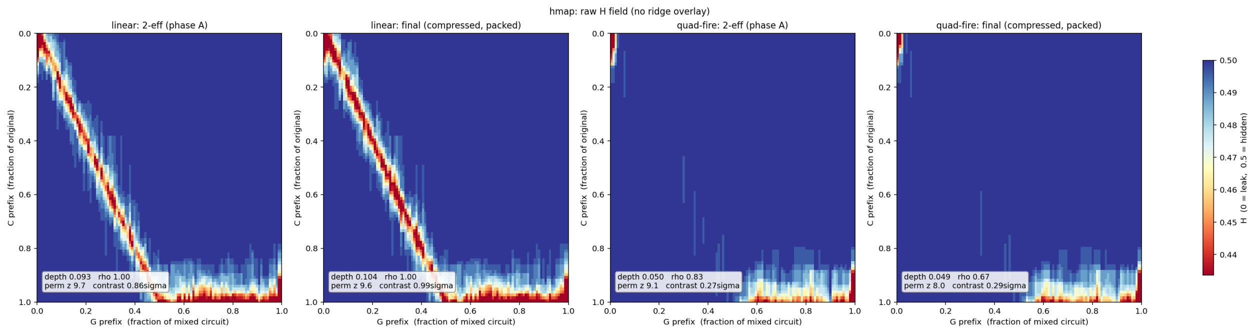
The figures in §5.1–§5.5 were read by **mean H** and by the count of **exposed rows** (segments with a cell below 0.25/0.35). Both are blind to a ridge made of *exact* affine relations at a few bits per cell, and the ridge statistic of `plot_hmap_ridge.py` (depth, median-row depth, ρ) had never been computed for the pipeline circuits. Computing it showed a full-strength diagonal in every mixed circuit of the linear-read design: the raw gadget has ≈500–800 exact relations (median-row depth 0.008 — the "depth ≈ 0.05" of §5.1 is the I/O fringe plus a faint trace), the 2-eff snapshot ≈6,000 (median-row depth 0.07, $\rho = 1.00$, permutation $z = 9.6$), and the compressed, packed final ≈7,000 (depth 0.10). Each relation is one C bit equal to the XOR of the data wire and its ∼6 linearised band wires: a **masked-read window**. The undo of a read is a write to the operand wire, pinned only by that wire's next read, so every reordering stage — the DB splice moves of phase A, fmix's final uniform float, the crossing walk — stretches the window to the operand's idle interval (isolated: reordering-only moves 503 → 601 in snapshots, the final float alone → 4,555; DB moves saturate by ≈1 move/gate). No parameter fixes it: extra filler LGIs or repair slots dilute it by $\approx \div 2$ at +35–55% gates, mask width is irrelevant, and `max_open`= 2 is worse (full tables in `RIDGE_QUADFIRE_20260906.md`).



Quad-fire (§2 step 2) removes the window instead of diluting it. Same recipe, 2-eff pipelines, exact relations per stage (depth / median-row depth):

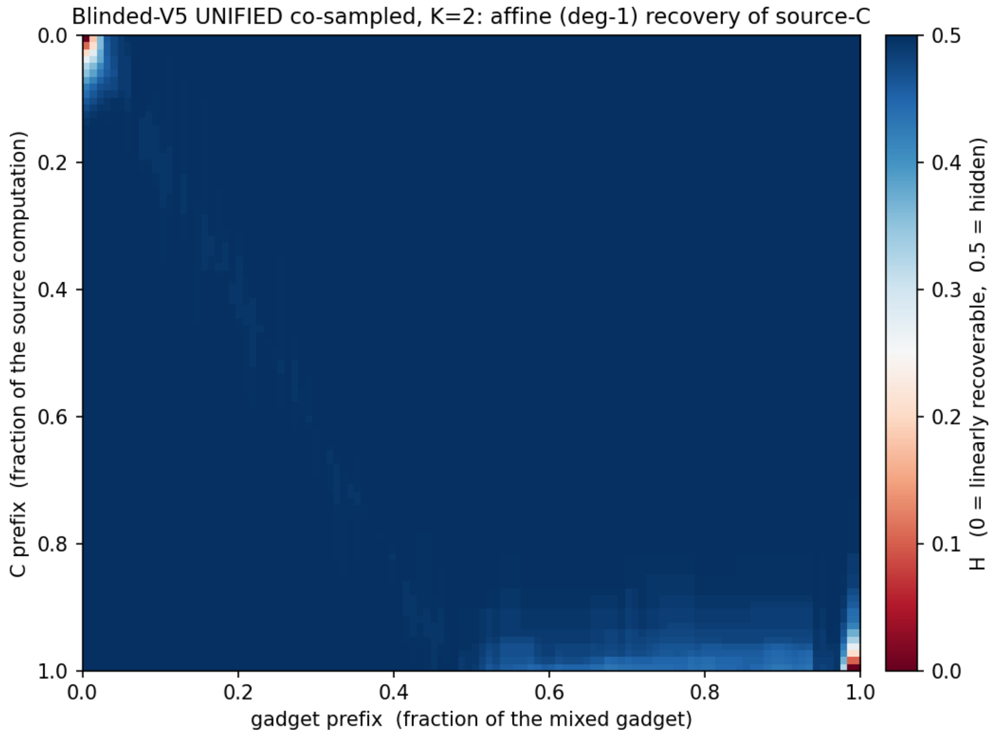
stage	linear read	quad-fire
gadget	563,356 g, 834 (0.049 / 0.008)	497,548 g, 500 (0.048 / 0.000)
2-eff phase A	962,429 g, 6,032 (0.093 / 0.066)	898,416 g, 443 (0.050 / 0.004)
split	1,839,248 g, 5,984 (0.096 / 0.070)	1,507,931 g, 478 (0.049 / 0.004)
crossing	3,345,382 g, 4,574 (0.081 / 0.051)	2,738,434 g, 368 (0.049 / 0.004)
final (fcompress + pack)	337,194 packed, 7,179 (0.104 / 0.082, ρ 1.00)	311,285 packed, 400 (0.049 / 0.004)





The residual 150–500 relations of quad-fire are single-wire identities in the delivered-output rows and at the raw input port — the public I/O fringe of §5.2 — and they do not grow with mixing (flat from the gadget to 3 moves/gate in phase A and through split, crossing and compression). The degree-2 adversary of §5.3 (products over a 24-data + 24-band slice) is likewise flat for quad-fire (141 $\to$ 102 after a float) where the linear read goes 481 $\to$ 4,978; a full degree-2 adversary with every band-pair product would cancel three open quadratic masks in either design (the existing sample-count-bounded posture), so quad-fire closes the affine channel, which was the open one.

5.1 Affine ridge — raw gadget, linear read

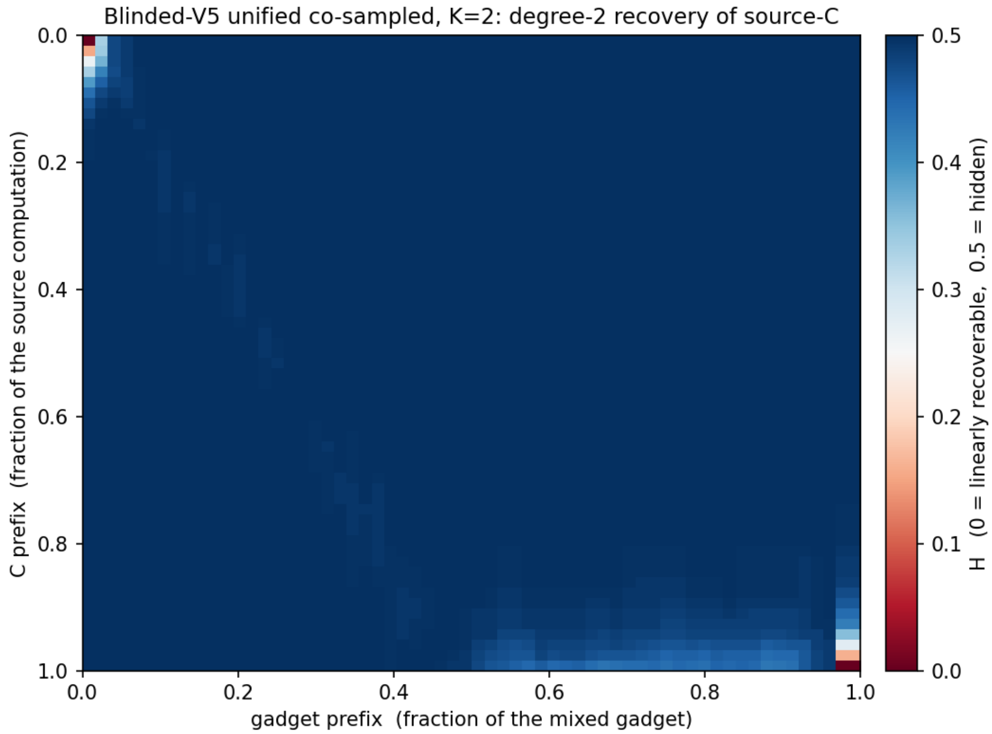


meanH ≈ 0.498, ridge depth ≈ 0.05; the interior ($\sim$78% of rows) is fully hidden ($H \approx 0.49$), and the only recovery is a thin fringe at the two endpoints.

5.2 The exposed fringe is public I/O — input segments are not leaked

The $\approx$6–10% of source-C segments that are linearly exposed (min-$H < 0.25$–0.35) are the two corners of the diagonal. The **input side is masked as soon as the compute begins** — recovery is confined to the extreme top-left corner, i.e. the raw input port where x is public *before* any masking. The output side is the necessarily-delivered $C(x)$. So the input wire segments show a **lack of linear exposure** beyond the trivial public input; the boundary-cushion knobs measured no effect because the fringe is unavoidable public I/O, not a leak.

5.3 Degree-2 recovery



meanH ≈ 0.497, depth ≈ 0.05. The quadratic adversary reaches only $\sim$0.01 deeper than affine, into the same low-degree I/O fringe; the interior stays fully hidden. Degree-2 is flat across K, like affine.

5.4 Rerand: bursts, straddle vs repair ($n_A = 256$, affine deg-1)

Each rerand slot is a **burst** of $F \approx 8\text{K}$ gates on one band wire; the **slot** count — not the gate count — drives the effect, since a straddle slot closes the masks reading b exactly once per slot. An early single-gate-dose study fixes where the thinning knee is (dose = total straddle/repair *gates*):

straddle	repair	gadget	meanH	affDepth
0	0	670k	0.4979	0.0550
1000	3000	668k	0.4959	0.0514
8000	0	607k	**0.4878**	**0.0681**
0	8000	780k	0.4985	0.0549

Heavy STRADDLE thins (meanH ↓, ridge ↑, gadget *shrinks* as masks close); heavy REPAIR does not (both = baseline, gadget *grows* from the pre/post pairs). The production plan keeps straddle **slots** on the safe side of the ≈ 1024 knee: auto $\approx m/(4K)$ ($\sim$875 for the half-size sandwich) $\times F = 8\text{K} \approx 2m$ band-refresh gates, **no repair**. Fewer, fatter slots mix the band as hard as many single updates while interfering less with the LGIs (and give the band its data-wire signature), and — because the thinning is driven by the *slot* count — they sit at the baseline floor (deg-1 meanH = 0.4966, deg-2 0.4967).

5.5 Through the GSS pipeline ($K = 2$ vs $K = 4$)

Four arms through `gss_mix.sh --gadgetization-mode blinded-v5` (n=128, half-size $|C| = |D| = 3000$, `--expand 2`, fresh independent CSPRNG seeds), measured at the **2-eff snapshot** (phase-A grow to ≈1.7×, an early mixing checkpoint; `hmap_affine --degree 1`, 92 C-segments):

arm	gadget g_{in}	2-eff gates	meanH	exp. < 0.25	< 0.35
K=2, hold 10	563,946	962,598	0.4945	6/92	11/92
K=2, hold 20	564,394	969,380	0.4942	6/92	11/92
K=2, hold 30	564,674	960,670	0.4951	6/92	9/92
K=4, hold 10	1,362,010	2,300,885	0.4965	6/92	10/92

- **meanH** ≈0.494–0.497 **for both** K — affine-neutral across K through the pipeline, matching the raw-gadget measurements (§5.1) and the pre-firing-fix baseline. (This is the corrected build with burst rerand + the masking floor + the even-filler / no-drain schedule; the pipeline statistics are unchanged, as the band-only refinements predict.)
- The exposed-C count (6/92 at < 0.25, ∼9–11/92 at < 0.35) is the public **I/O boundary** of §5.2, confirmed by the per-segment profile: min-H ≈0 only at the input port (raw x) and output port ($C(x)$), climbing onto the hidden plateau within ∼5% of C.
- The three K=2 arms agree because the 2-eff snapshot **precedes the hold phase**, so it is hold-independent — a seed-level consistency check.

Mean H and the exposed-row count do not see the ridge: these same 2-eff snapshots carry ≈6,000 exact relations (depth 0.09–0.10, $\rho = 1.00$; §5.0). The arms in this table use the linear read; quad-fire arms are the follow-up. End-of-pipeline numbers for the linear read (post split / crossing / fcompress) are in §5.0's table (2-eff pipelines) and in `RIDGE_QUADFIRE_20260906.md`.

5.6 Linear correlation with C's firing predicates; idle-bare intervals (2026-09-06)

The affine measures above are *exact* GF(2) tests; they are blind to a wire that is merely *statistically* close to a source value. `fire_corr` (`red_team_tests/bin/leakage/fire_corr.rs`) measures the phi (Pearson) correlation between each C gate's **firing predicate** ($comp \oplus \bigwedge lit$, over 4096 random inputs) — or, with `--c-segments`, each C **state bit** — and every G segment and every G-gate increment, against a shuffled null (null max ≈0.10). Calibration: a fire vs its *bare* operand is phi 0.577; vs the operand under one **g57** mask term 0.289 (the term $1 \oplus \neg x \wedge y$ is 1 with probability 3/4); two terms 0.144. Read on the raw gadget ($n = 128$, same C and seeds throughout; "interior" = C gates in the middle 70% of C, which excludes the public I/O fringe; counts out of 3000 C gates, 2101 interior):

build	gates	interior idle-bare intervals	fires: int. median phi	≥ 0.3	≥ 0.5	state bits: int. ≥ 0.5	exact copies
quad-fire, before the coverage fixes (0fb96997)	496,386	448 (med. 11k)	0.289	907	19	336	1
balanced, before the fixes	939,214	735 (med. 19k)	0.079	98	67	38	13
quad-fire + coverage fixes	514,502	**0**	0.287	907	15	315	0
balanced + coverage fixes (default)	984,384	**0**	**0.078**	**31**	15	**9**	0
null (shuffled)			0.077	0	0	0	0

Three findings.

1. **The one-mask population.** Half of C's gates have a segment on an operand wire at phi $\approx$0.29 = the operand under a *single* biased mask term: masking depth between reads is typically 1–2 terms, and each term is biased. This is the channel `balanced` closes — the interior median drops from 0.29 to the null floor (0.078) and the state-bit hits from 315 to 9 — at +92% gates. It survives the pipeline unchanged (quad-fire 2-eff: 862 interior ≥ 0.3; final 642), because reordering does not change a wire's statistics.

2. **Idle-bare intervals (a build defect, fixed).** The strongest interior hits (phi $\approx$0.58 = *bare* operand) were exact copies of C state bits sitting on a wire for thousands of gates. A rerand burst on band wire b closes every open LGI reading b; when that was the wire's *only* open LGI the wire was left bare until its next filler open — every interior bare interval of the earlier builds (448 / 735, all of them) had this trigger, and balanced LGIs (three band wires instead of two) were hit 50% more often. The fix is the *cover-replacement* rule (§2: open a fresh LGI on the wire before the close; $\approx$430 (plain) / $\approx$780 (balanced) replacements per build, $\approx$0.3% gates). A second, smaller population were seldom-used (high-half) wires whose only masks were read-time top-ups, undone after each fire; the *read-cover* rule keeps the first top-up open ($\sim$115–125 per build, no cost). A third, $\sim$1 per build, was two identical open pairs cancelling; LGI sampling now avoids a wire's open pairs. After the three rules the census finds **no** interior bare interval in either variant, and the remaining ≥ 0.5 hits (15, identical in both variants) sit at the ends of the interior window next to the public I/O fringe. The pre-fix defect is in every earlier blinded-V5 deliverable (incl. the K=2 2-eff files); reordering stretches those bare windows (interior ≥ 0.5: gadget 19 $\rightarrow$ 2-eff 81 $\rightarrow$ final 52).

3. **What the scans do not see.** The band itself is nearly uniform after the bursts ($|P - \frac{1}{2}| \approx 0.02$), the AND seed ($x_i \wedge \neg x_j$, bias $\frac{1}{4}$) versus the balanced XOR seed makes no difference to the residual, and restricting bursts to band-only controls (`BV5_BURST_BANDONLY=1`) does not help either — the defect was structural, not statistical.

Diagnostics: `fire_corr --probe <C gate list>` prints, for each listed C gate, its best fire-partner segment's phi against the gate's $(a, b, c_{\text{old}}, \textit{fire}, c_{\text{new}})$ (the gauntlet's 5-tuple), which is what identified the bare copies (a: +1.000); `BV5_DIAG=1` reports the replacement and read-cover counts.

5.7 The gadget gauntlet (colleague's trace-and-audit battery), $n = 32/64$ (2026-09-07)

`tests/gauntlet` records, for random inputs, every wire's initial value and every gate's flip and new target value of a gadgetized r57 chain, and runs six attacks against the chain's $(a, b, c_{\text{old}}, \textit{fire}, c_{\text{new}})$

per source gate: `a1` (a feature equals a target), `xrows` (a target is affine in one prefix's wire state), `xtrace` (affine in the whole trace), and `w1/w2/w3` (correlation with one feature, or an XOR/AND/OR/AND-NOT of two or three, capped strided subsets), flagged above the NULL column's maximum and 6σ. Blinded-V5 enters as native arms (`blindedv5` = plain masks, `blindedv5_balanced`): production preset, **encoded I/O** (`encoded_io`: `max_open` masks per wire pre-opened and the drain skipped, the encode/decode gates applied out of band — the convention of the colleague's share-encoded arms, so no raw input or output is a wire value), band $\max(32, n)$ wires starting uniformly random. The harness's own controls (`none`, `secretshare14`, `bandproduct92`) light up every attack. Numbers are nontrivial hits / flags out of $5k$ targets (k source gates):

n	k	arm	mix	a1	xrows	xtrace	w1	w2	w3
32	64	balanced	off	0	0	320	0	0	0
32	256	balanced	off	0	0	1280	0	0	0
64	64	balanced	off	0	0	320	0	0	0
64	256	balanced	off	0	0	1280	0	0	0
32	64	quad-fire (plain)	off	0	0	320	320	27	35
64	256	quad-fire (plain)	off	0	0	1280	1280	5	5
64	256	`none` control	off	1096	840	1096	1280	1214	442
32	64	balanced	on	0	1	320	28	0	0
32	256	balanced	on	0	0	skip	47	0	0
64	64	balanced	on	0	0	320	0	0	0
64	256	balanced	on	0	0	skip	60	0	0
64	256	quad-fire (plain)	on	7	7	1280	1255	23	13

Reading it:

- **Unmixed, the balanced build passes every discriminating attack** at both sizes and both chain lengths; the plain build fails `w1` on every target (the one-mask bias of §5.6) and `w2/w3` on a few. `xtrace` is an identity for XOR masking — every mask term is the flip of the gate that applied it, so each plaintext is an exact XOR of trace features — and fires for every XOR-based arm including the harness's controls; only the colleague's share-encoded `nonlinear193` passes it (nonlinear decode). It is not a statement about what an adversary without the targets can do.

- **The harness found three more generator gaps**, each a sampling coincidence that made a mask sum biased: two open masks on a wire sharing a pair, a balancing wire, or a balancing wire equal to another mask's pair wire ($x \oplus \neg x \wedge y = x \vee y$); and, at a fire, a band wire shared between the target's masks and the operands' polynomials (phi 0.1–0.25 segments mid-fire). The disjoint-sampling rule (§2) and the fire-cover bracket (§2, step 4) closed them; the bracket costs ≈3% gates.

- **Mixed cells are the harness's own mixer** (store-free crossings, copy splits and conjugation twists at 20k moves), not the pipeline: its splits break the quadratic masks into CNOT-sized pieces and reorder them, so intermediate states carry partial masks. That produces sporadic `w1` flags on the balanced arm (max phi 0.4–0.5; none at $n = 64$, $k = 64$) and, on the plain arm, exact plaintext copies (`a1` 7 at $n = 64$, $k = 256$). The pipeline's own reorderings are measured end to end with `fire_corr` instead (§5.6).

- At $n = 8$ (the colleague's default) the picture is the same, with the encoded-I/O convention the only way to make the comparison meaningful: with plaintext I/O nearly every target is an input or output at that size.

$n = 256$ **(V5 input width 256, band 256), balanced, unmixed**, `max_open` 3 versus the size-neutral `max_open` 2 (`blindedv5_balanced_mo2`):

k	`max_open`	a1	xrows	w1	w2	w3	note
256	3	0	0	0	0	0	
512	3	0	0	0	0	0	
256	2	0	0	$4 \to 0$	6	0	w1 before $\to$ after the burst-control rule
512	2	0	0	0	$20 \to 0$	0	w2 before $\to$ after the burst-control rule

Two more things came out of it. (i) A **burst gate** whose controls are a masked data wire and that wire's own balancing wire strips the mask's uniform term ($(x \oplus z \oplus q) \wedge \neg z$): the burst's flip correlated with the plaintext at phi 0.14–0.26 — the sixth and last sampling rule (§2: burst controls avoid a data control's mask wires). (ii) **`max_open` 2 is not equivalent to 3.** With a single open balanced mask $x \oplus z \oplus q$, the XOR of the wire's segment with any visible monomial that contains z (a fire monomial or a burst reading z) cancels the uniform term and leaves the biased quadratic part (phi 0.23); the harness's capped two-feature scan finds it when its strided subset happens to hold such a pair (w2 6 of 1280 at $k = 256$; 0 at $k = 512$ after the burst rule) and its mixer wrecks it (a1 10, xrows 80, w1 201 at $k = 256$ mixed, versus w1 6 / w2 1 for `max_open` 3). Three open masks carry more independent uniform terms than a pair of features can cancel. So the +10% variant buys the w1 result but not the multi-feature one; `max_open` 3 at +92% is the clean build.

5.8 Two-control emission of the fire, and the cost of it (2026-09-08)

The store the mixing stages splice from is a complete ball of `g57` gates: one target, **two** controls. A gate with three or more controls matches nothing in it, so no splice ever touches it — it passes through phase A, the split and the crossing walk as a fixed object, carrying the monomial structure of C's gate verbatim. The first quad-fire emission expanded each fire into monomials of degree up to 4 and so violated this badly:

build ($n = 128$, same C)	0 ctrl	1 ctrl	2 ctrl	3 ctrl	4 ctrl
wide monomials (0d81a9f3)	0.8%	14.3%	47.8%	**30.6%**	**6.5%**
scratch-wire product (interim, rejected)	0.5%	26.1%	73.2%	0.2%	0
borrowed dirty wires (current)	0.28%	6.25%	**93.39%**	**0.08%**	**0**

Clean ancillas were the wrong way to do it. The interim version appended four clean scratch wires. Measured: some non-data wire was identically 0 at **89.5%** of gate positions, a *function-level* invariant that survives any equivalent rewriting and whose non-zero stretches delimit exactly the fire blocks — the segmentation that hidden firing (§3) exists to prevent. It also silently required the evaluator to zero them: starting them at 1 changed the payload output in 4,062 of 8,192 bit positions, so the circuit computed garbage without failing. With borrowed dirty wires the same statistic is 0.2%, and that residue is only the band before module 2 has seeded it.

Where the gates go. Per fire, over 7,920 fires at $n = 128$ (235 gates each, of which 137 write the target and 93 are ancilla borrows):

term class	gates per fire	share
linear × quadratic (4-gate bracket)	111.2	47%
quadratic × quadratic (8-gate bracket)	55.6	24%
linear × linear (1 gate)	27.8	12%
wire × quadratic (4-gate bracket)	20.1	9%
wire × linear (1 gate)	10.1	4%
constant × everything	8.5	4%
$a'b'$, constant flip	1.6	1%

Each operand carries ≈2.6 open masks; each mask $1 \oplus y \oplus x{\cdot}y \oplus z$ contributes **two** linear terms (y from the `g57` form, z from the balancing CNOT) and **one** quadratic term. An operand is therefore ≈10 atoms and the product ≈100 terms; terms with one quadratic factor cost 4 gates and with two cost 8, which is why the two quadratic classes are 71% of the fire. Levers, should size matter: `max_open` 2 (≈130 gates per fire, but weaker against the two-feature scan, §5.7), or a mask with no linear y term (≈146, at the cost of `g57`-digestibility). `BV5_DIAG=1` prints this census.

Measurements at $n = 128$ (2,024,834 gates, 512 wires): forward and reverse verify pass, the exhaustive test passes in every configuration, `fire_corr` interior median 0.080 with 31 gates ≥ 0.3 and 23 state bits ≥ 0.3 / $9 \geq 0.5$ — unchanged from the wide-monomial build — 0 affine relations, 0 bare intervals, 0.0% of covered wire-time at a single open mask, and the hot manifest lists exactly the 512 I/O fringe segments.

What the requirement costs. Over five gauntlet instances ($n = 64$, $k = 64 \ldots 68$, 16,384 correlation samples) the current build shows w3 flags on 3 and 1 targets in two of the five, everything else zero; the wide-monomial control is clean on all five. The residual is weak (phi ≈0.065 against a 0.047 threshold) and is plausibly inherent rather than a bug: building a degree-4 monomial from 2-control gates must materialise the degree-2 partial products as gate **flips**, and a flip *is* the product, so unlike a wire value it cannot be blinded. The trade is deliberate — wide monomials pass this scan but leave 37% of the gadget unspliceable, which is the larger exposure.

6 Tradeoffs and the current choices

- **Mask width K vs size.** Larger K widens each mask, not the temporal length; hiding is flat in K while size grows ∼linearly and the read cost quadratically, so **K=2**; K=4 only for a wider read-mask.
- **Band turnover vs size (bursts, slots vs F).** More band updates = more entropy and stronger resistance to commutation-back, but STRADDLE **slots** thin the masking past ≈ 1024 and REPAIR slots grow the gadget. Bursting decouples the two knobs: F (gates/slot) sets how hard each band wire is stirred and how data-like its activity looks, while the **slot count** (≈ $m/4K$, held under the knee) sets the thinning. Auto $m/4K$ slots $\times F$ = 8K, no repair, sits at the baseline floor.
- **Masking floor is a cheap hedge.** `min_mask` guarantees every operand is read under ≥ 3 masking wires (worst-case $|\rho|$ $2 \to 4$; mean unchanged); it fires only on the rare under-floor draw, costing ∼0.2% gates — a hedge against a low-probability thin-masking event, not a routine cost.

- **Firing-hiding is free** in the co-sampled build (it reuses the existing LGI budget); earlier fixed-scaffold approaches paid coverage (23–70%) or +43% size.
- **The residual fringe is I/O, not a knob** — no parameter removes the public input/output boundary.
- **Coverage is enforced, not budgeted.** The u_w+1 budget sets the mask *statistics*; the six coverage rules (§2, §5.6: cover-replacement across a burst, read-cover on an uncovered operand, disjoint mask wires incl. duplicate-free pairs, the fire-cover bracket, the burst-control rule) make "no wire is ever bare between its first and last LGI" a property of the build rather than a likely outcome, for ≈3.5% gates (the bracket is ≈3% of it); `min_open` 2 lifts the floor from one mask to two for +0.03%.
- **Balanced masks (default since 2026-09-07).** The biased `g57` term leaves a wire under one mask linearly correlated with its plaintext (phi 0.29); `balanced` removes it (null floor) for +92% gates at K=2/`max_open` 3 and is the build that passes the gauntlet's correlation battery (§5.7). The exact-affine measures are unaffected either way, and the cost is the read polynomial, $(3\,\texttt{max\_open} + 2)^2 = 121$ monomials per fire instead of $(2\,\texttt{max\_open} + 2)^2 = 64$. `max_open`= 2 **balanced** brings the polynomial back to 8×8 = 64: measured 565,374 gates at n = 128 (+10% over the plain `max_open` 3 build's 514,502, versus +92% for balanced `max_open` 3's 984,384), with the same `fire_corr` statistics as balanced `max_open` 3 (interior median 0.076 vs 0.078, p90 0.137 vs 0.133, no bare intervals) — but the gauntlet's two-feature scan separates them: one open mask is one uniform term, which a single visible monomial cancels (§5.7). `max_open` 3 remains the clean build.
- **Gate width is a hard constraint, not a cost knob (§5.8).** The mixing store is a `g57` 2-control identity ball, so any gate with 3+ controls is never spliced and survives the pipeline carrying C's structure. The fire is emitted through scratch wires as 2-control gates rather than as wide monomials, for +12% gates and no change to any leakage measure. The only 3-control gates left in a full gadget are ≈0.2%, from the sandwich's balanced junk-guard.
- **Read mode: quad-fire, not linearisation.** Linearising a read is the cheapest way to fire (degree-2 monomials only) but leaves the operand affine in band wires for a window that reordering stretches into the pipeline ridge (§5.0). Firing from inside the quadratic masks costs degree ≤ 4 monomials, needs no linearise/undo bracket, makes the gadget ∼12% smaller, and keeps the ridge at the fringe end to end. Repair slots are counter-productive with quad-fire (they close masks around bursts); keep `max_open`= 3.